

PRACTICE EXERCISES

Complex Real-World Scenarios

Data Modeling Intensive Training

DAY 1 EXERCISES: ERD Design

Exercise 1: Complete Hospital Management System

Time Allocation: 8:30 AM - 12:00 PM (3.5 hours)

Business Context

You are hired as a database designer for a large multi-specialty hospital that wants to digitize its entire operations. The hospital currently uses paper-based records and spreadsheets, leading to inefficiencies and errors. Your task is to design a comprehensive database that will manage all aspects of hospital operations.

Detailed Requirements

1. Patient Management:

- Patients register with the hospital providing: Patient ID (auto-generated), full name (first, middle, last), date of birth, gender, blood type, marital status
- Contact information: Multiple phone numbers (home, mobile, work), primary email, emergency contact details
- Address details: Street, apartment/house number, city, state, postal code, country
- Insurance information: Insurance company name, policy number, group number, coverage percentage, expiry date
- Medical history: Known allergies (multiple), chronic conditions, previous surgeries
- Emergency contacts: Name, relationship to patient, phone number (multiple contacts per patient allowed)

2. Medical Staff Management:

- Doctors: License number (unique), full name, specialization (can have multiple), date of hire, phone, email, office location
- Nurses: Employee ID, name, qualification level (RN, LPN, CNA), shift preference, phone, email
- Support staff: Technicians, receptionists, pharmacists - track ID, name, role, department, shift
- All staff have one primary department but doctors can consult across departments

3. Department Structure:

- Departments: Code (unique), name, building, floor, phone extension, operating hours
- Each department has one head doctor (must be a doctor in that department)
- Track when the head doctor assumed the position (date appointed)
- Departments: Cardiology, Neurology, Orthopedics, Pediatrics, Emergency, ICU, Surgery, Radiology, Laboratory, Pharmacy

4. Appointment System:

- Patients schedule appointments with specific doctors

- Appointment details: Unique appointment ID, date, time slot, status (scheduled/confirmed/completed/cancelled/no-show)
- Reason for visit, appointment type (consultation/follow-up/procedure)
- Room number assigned for appointment, estimated duration
- Actual check-in time and check-out time (may differ from scheduled)
- Doctor's notes after appointment (diagnosis, observations)

5. Medical Records & Diagnosis:

- For each appointment, record diagnosis codes (ICD-10), diagnosis description, severity level
- Vital signs recorded during visit: temperature, blood pressure, heart rate, respiratory rate, oxygen saturation, height, weight
- Clinical notes: Doctor's detailed observations and examination findings
- If patient's medical record is deleted, all related vital signs and clinical notes should also be deleted

6. Prescription Management:

- Doctors prescribe medications during appointments
- Medication details: Generic name, brand name, dosage strength, form (tablet/capsule/syrup/injection)
- Prescription details: Dosage instructions, frequency, duration, total quantity
- Special instructions (take with food, avoid alcohol, etc.)
- Refill allowed (yes/no), number of refills
- Pharmacy fulfillment: date dispensed, pharmacist ID who dispensed

7. Laboratory Tests:

- Doctors order lab tests for patients
- Test types: Blood tests, urine tests, X-rays, MRI, CT scan, ECG, etc.
- Each test type has: Code, name, description, typical cost, sample type required
- Test orders track: Order ID, date ordered, priority (routine/urgent/stat), special instructions
- Sample collection: Date/time collected, collected by (staff member), sample ID
- Test results: Date/time completed, result values, normal range, interpretation, technician who performed test
- Results reviewed by doctor: review date, doctor's comments
- Tests are performed in specific labs (Hematology Lab, Radiology Lab, Microbiology Lab, etc.)

8. Hospital Admission & Discharge:

- Patients may be admitted to hospital for treatment
- Admission details: Admission ID, admission date/time, admission type (emergency/elective/transfer)
- Admitting diagnosis, attending doctor, expected length of stay
- Room assignment: Building, floor, room number, bed number
- Room type: Private, semi-private, ward, ICU
- Daily nursing notes during admission
- Discharge information: Discharge date/time, discharge type (routine/against medical advice/transferred/deceased)

- Discharge diagnosis (may differ from admission), discharge instructions, follow-up appointments required

9. Surgery & Procedures:

- Track surgical procedures performed on patients
- Procedure details: Procedure code, name, description, typical duration
- Surgery schedule: Date, start time, end time, operating room number
- Primary surgeon and assisting surgeons (multiple)
- Anesthesiologist assigned, anesthesia type used
- Nursing staff present during surgery
- Surgery outcome notes, complications if any, post-op instructions

10. Billing & Payments:

- Generate bills for: Consultations, procedures, lab tests, medications, room charges
- Bill details: Bill ID, date generated, due date, total amount, tax amount
- Itemized charges: Each service/item with quantity, unit price, total
- Insurance coverage applied: Amount covered by insurance, patient responsibility
- Payment tracking: Payment date, amount paid, payment method (cash/card/insurance/check)
- Transaction reference number, outstanding balance

Required Deliverables

1. Entity Identification Document

List all entities with classification (strong/weak). Expected: 18-25 entities.

2. Attribute Analysis Document

For each entity, list all attributes and classify as: Simple, Composite, Multi-valued, Derived, or Complex. Mark primary keys and partial keys.

3. Relationship Documentation

List all relationships with: Degree (unary/binary/ternary), Cardinality (1:1, 1:N, M:N), Participation (total/partial for each entity).

4. Complete ERD

Draw using Crow's Foot or Chen notation. Ensure all entities, attributes, relationships, cardinality, and participation constraints are clearly shown.

5. Assumptions Document

List all assumptions made where requirements were ambiguous. Justify your decisions.

Design Challenges to Consider

- Emergency Contact: Strong or weak entity? Why?
- Should patient age be stored or derived?
- How to handle a doctor who is head of multiple departments?
- Should vital signs be attributes or separate entity?
- Where should prescription fulfillment information be stored?
- How to model the surgery team (surgeon, anesthesiologist, nurses)?

Exercise 2: Multi-Tenant E-Commerce Platform

Time Allocation: 1:00 PM - 3:15 PM (2.25 hours)

Business Context

Design a database for a marketplace platform similar to Amazon or eBay where multiple vendors can sell products to customers. The platform charges commission on sales and provides various services to both buyers and sellers.

Detailed Requirements

1. User Management (Buyers & Sellers):

- Users register with: User ID, username (unique), email (unique), password hash, full name (first, last)
- Registration date, last login timestamp, account status (active/suspended/closed)
- User type: Some users are only buyers, some only sellers, some both
- Profile information: Profile picture URL, bio, date of birth, phone number
- Sellers have additional info: Business name, tax ID, business address, business phone, business email, seller rating
- Sellers must be verified (verification status, verification date, verified by admin ID)

2. Address Management:

- Users can save multiple shipping addresses
- Address details: Address type (home/office/other), recipient name, street address, apartment/suite, city, state/province, postal code, country, phone
- One address marked as default shipping address
- If user is deleted, all their addresses should be deleted

3. Payment Methods:

- Users save payment methods: Credit/debit card, PayPal, bank account
- Card info: Card type (Visa/Mastercard/Amex), last 4 digits only, cardholder name, expiry month/year
- Billing address associated with each payment method
- One payment method marked as default, is_verified flag

4. Product Catalog:

- Sellers create product listings
- Product details: Product ID, title, detailed description, brand, model number
- Pricing: Base price, sale price (if on sale), currency, cost price (for seller only)
- Inventory: Quantity in stock, SKU, warehouse location, reorder level, reorder quantity
- Product dimensions: Length, width, height, weight (for shipping calculations)
- Product status: Draft, active, out of stock, discontinued
- Date listed, date last updated, number of views, number of sales
- Product can have variants (e.g., different sizes, colors) - each variant has its own SKU, price, inventory

5. Product Images:

- Products have multiple images
- Image details: Image URL, alt text, display order (for sorting), is_primary flag
- Date uploaded, image size (for optimization)

6. Category Hierarchy:

- Products organized in hierarchical categories
- Category: Category ID, name, description, image URL
- Categories can have parent categories (e.g., Electronics > Computers > Laptops)
- A product can belong to multiple categories
- Each category has attributes specific to products in that category (e.g., screen size for laptops)

7. Shopping Cart:

- Buyers add products to shopping cart
- Cart items: Product, quantity, date added, variant selected (if applicable)
- Save for later feature: items can be moved to saved list
- Cart persists even if user logs out

8. Orders & Checkout:

- Buyers create orders from cart items
- Order details: Order number, order date, order status (pending/processing/shipped/delivered/cancelled/refunded)
- Shipping address (snapshot at order time), billing address
- Subtotal, shipping cost, tax amount, discount applied, total amount
- Payment method used, transaction ID from payment gateway
- Order items: For each item, store product snapshot (price at purchase time, title, seller)
- Quantity, unit price, discount applied, total price for that item
- Items from different sellers in one order should create separate shipments

9. Shipping & Fulfillment:

- Track shipment details for each order
- Shipping carrier (UPS, FedEx, USPS, DHL), service type (standard/express/overnight)
- Tracking number, shipping date, estimated delivery date, actual delivery date
- Shipment status (pending/picked up/in transit/out for delivery/delivered/failed)
- Delivery confirmation: Date/time, signature required, recipient name, delivery notes

10. Reviews & Ratings:

- Buyers can review products they purchased
- Review details: Rating (1-5 stars), review title, detailed review text, date posted
- Verified purchase flag (only if bought through platform)
- Helpful votes from other users (like/dislike)
- Seller can respond to reviews with comments
- Review images (buyers can upload photos)

11. Wishlist & Favorites:

- Users can add products to wishlist
- Track date added, notify when price drops, priority level
- Users can create multiple wishlists (Birthday, Holiday, etc.)
- Wishlists can be shared (public/private/shared with specific users)

12. Promotions & Discounts:

- Sellers create promotional campaigns
- Promotion details: Promo code, description, discount type (percentage/fixed amount)
- Discount value, minimum purchase amount, maximum discount cap
- Valid from date, expiry date, usage limit per customer, total usage limit
- Applicable to specific products or categories
- Track how many times each promo code has been used

13. Returns & Refunds:

- Customers can request returns for orders
- Return request: Return ID, order reference, items being returned, reason code, detailed reason
- Request date, status (pending/approved/rejected/completed)
- Return shipping details: Carrier, tracking number, return received date
- Inspection notes, refund amount, refund date, refund method

Critical Design Questions

6. How do you handle users who are both buyers and sellers?
7. Should shopping cart be an entity or a relationship? Justify.
8. Why must product price be stored in the order item when it already exists in product?
9. How to model the hierarchical category structure?
10. How to ensure reviews are only from verified purchases?
11. What happens to seller's products if seller account is deleted?

Deliverables (Same as Exercise 1)

12. Entity identification document
13. Complete attribute analysis
14. Relationship documentation
15. Complete ERD
16. Assumptions document

Exercises for Day 2, Day 3, and Day 4 continue...

Full document includes detailed specifications for:

- *Day 2: Mapping Exercises (Hospital & University Systems)*
- *Day 3: Normalization Exercises (Sales & Student Records)*